响应示例

Element Software

NetApp
January 17, 2022

This PDF was generated from https://docs.netapp.com/zh-cn/element-software/api/reference_element_api_response_example_getconfig.html on January 17, 2022. Always check docs.netapp.com for the latest.

目录

响应示例 1
- 了解更多信息 1
- getconfig 1
- GetClusterHardwareInfo 4
- GetLldpInfo 17
- GetNetworkConfig 47
- GetNodeHardwareInfo （ iSCSI 的输出） 51
- GetNodeHardwareInfo （光纤通道节点的输出） 53
- GetNvramInfo 60
- ListActiveNodes 69
- ListActiveVolumes 72
- TestHardwareConfig 82

响应示例

此处提供了完整的响应示例。

- getconfig
- GetClusterHardwareInfo
- GetLldpInfo
- GetNetworkConfig
- GetNodeHardwareInfo （ iSCSI 的输出）
- GetNodeHardwareInfo （光纤通道节点的输出）
- GetNvramInfo
- ListActiveNodes
- ListActiveVolumes
- TestHardwareConfig

了解更多信息

- "NetApp SolidFire 资源页面"
- "早期版本的 NetApp SolidFire 和 Element 产品的文档"

getconfig

`getconfig` 方法返回类似于以下示例的响应。由于篇幅限制，此响应仅包含集群中一个节点的信息。


```
{
    "id": 1,
    "result": {
        "config": {
            "cluster": {
                "cipi": "Bond10G",
                "cluster": "AutoTest2-Fjqt",
                "encryptionCapable": true,
                "ensemble": [
                    "1:10.1.1.0",
                    "3:10.1.1.0",
                    "4:10.1.1.0"
                ],
                "mipi": "Bond1G",
                "name": "NLABP2605",
                "nodeID": 1,
                "pendingNodeID": 0,
```

```
                "role": "Storage",
                "sipi": "Bond10G",
                "state": "Active",
                "version": "11.0"
            },
            "network": {
            "Bond10G": {
                "#default": false,
                "address": "10.1.1.0",
                "auto": true,
                "bond-downdelay": "0",
                "bond-fail_over_mac": "None",
                "bond-miimon": "100",
                "bond-mode": "ActivePassive",
                "bond-primary_reselect": "Failure",
                "bond-slaves": "eth0 eth1",
                "bond-updelay": "200",
                "dns-nameservers": "10.1.1.0, 10.1.1.0",
                "dns-search": "ten.test.company.net., company.net.",
                "family": "inet",
                "gateway": "10.1.1.0",
                "linkSpeed": 10000,
                "macAddress": "c8:1f:66:ee:59:b9",
                "macAddressPermanent": "00:00:00:00:00:00",
                "method": "static",
                "mtu": "9000",
                "netmask": "255.255.240.0",
                "network": "10.1.1.0",
                "physical": {
                    "address": "10.1.1.0",
                    "macAddress": "c8:1f:66:ee:59:b9",
                    "macAddressPermanent": "00:00:00:00:00:00",
                    "mtu": "9000",
                    "netmask": "255.255.240.0",
                    "network": "10.1.1.0",
                    "upAndRunning": true
                },
                "routes": [],
                "status": "UpAndRunning",
                "symmetricRouteRules": [
                    "ip route add 10.1.1.1/20 dev Bond1G src 10.1.2.2
table Bond1G",
                    "ip rule add from 10.1.1.1 table Bond1G",
                    "ip route add default via 10.1.1.254"
                ],
                "upAndRunning": true,
```

```
                "virtualNetworkTag": "0"
            },
            "eth0": {
                "auto": true,
                "bond-master": "Bond10G",
                "family": "inet",
                "linkSpeed": 10000,
                "macAddress": "c8:1f:66:ee:59:b9",
                "macAddressPermanent": "c8:1f:66:ee:59:b9",
                "method": "bond",
                "physical": {
                    "address": "0.0.0.0",
                    "macAddress": "c8:1f:66:ee:59:b9",
                    "macAddressPermanent": "c8:1f:66:ee:59:b9",
                    "netmask": "N/A",
                    "network": "N/A",
                    "upAndRunning": true
                },
                "status": "UpAndRunning",
                "upAndRunning": true
            },
            "lo": {
                "auto": true,
                "family": "inet",
                "linkSpeed": 0,
                "macAddress": "00:00:00:00:00:00",
                "macAddressPermanent": "00:00:00:00:00:00",
                "method": "loopback",
                "physical": {
                    "address": "0.0.0.0",
                    "macAddress": "00:00:00:00:00:00",
                    "macAddressPermanent": "00:00:00:00:00:00",
                    "netmask": "N/A",
                    "network": "N/A",
                    "upAndRunning": true
                },
                "status": "UpAndRunning",
                "upAndRunning": true
            }
        }
        }
    }
}
```

GetClusterHardwareInfo

`GetClusterHardwareInfo` 方法返回类似于以下示例的响应。


```
{
 "id": null,
  "result": {
   "clusterHardwareInfo": {
     "drives": {
      "1": {
        "description": "ATA      Drive",
        "dev": "8:0",
        "devpath": "/dev/disk/by-id/scsi-SATA_VRFSD3400GNCVMT205121562-
part4",
        "driveSecurityAtMaximum": false,
        "driveSecurityFrozen": true,
        "driveSecurityLocked": false,
        "logicalname": "/dev/sda",
        "product": "VRFSD3400GNCVMTJS1",
        "securityFeatureEnabled": false,
        "securityFeatureSupported": true,
        "serial": "205121562",
        "size": 299988156416,
        "uuid": "febe39ae-4984-edc0-e3a7-3c47608cface",
        "version": "515ABBF0"
      },
      "2": {...
      },
      "3": {...
      },
      "4": {...
      },
      "5": {...
      },
      "6": {...
      },
         .
         .
         .
    "44": {...
      }
      },
"nodes":{
  "1":{                       Storage Node
    "core_DMI:0200": {
```

```
    "description": "Motherboard",
    "physid": "0",
    "vendor": "SolidFire"
  },
    "fiber:0_PCI:0000:04:00.0": {
      "businfo": "pci@0000:04:00.0",
      "clock": "33000000",
      "description": "Fibre Channel",
      "physid": "0",
      "product": "ISP8324-based 16Gb Fibre Channel to PCI Express
Adapter",
      "vendor": "QLogic Corp.",
      "version": "02",
      "width": "64"
  },
    "Repeat fiber information": {...}
    "Repeat fiber": {...},
    "Repeat fiber": {...},
   }
 },
   "fans": {
     "Fan1A RPM": {
     "baseUnit": "RPM",
     "threshold": 840,
     "value": 4800
 },
     "Fan1B RPM": {...},
         .
         .
         .
     "Fan7B RPM": {...
     },
     "fibreChannelPorts": [
       {
        "firmware": "7.04.00 (d0d5)",
        "hbaPort": 1,
        "model": "QLE2672",
        "nPortID": "0x110c36",
        "pciSlot": 3,
        "serial": "BFE1341E09329",
        "speed": "8 Gbit",
        "state": "Online",
        "switchWwn": "20:01:00:2a:6a:a0:25:01",
        "wwnn": "5f:47:ac:c8:82:23:e0:00",
        "wwpn": "5f:47:ac:c0:82:23:e0:02"
       },
```

```
        {
         "firmware": "7.04.00 (d0d5)", {...}
         "firmware": "7.04.00 (d0d5)", {...}
         "firmware": "7.04.00 (d0d5)", {...}
        }
      ],
      "hardwareConfig": {
        "BIOS_REVISION": {
         "Passed": true,
         "actual": "1.1",
         "comparator": ">=",
         "expected": "1.0"
        },
        "BIOS_VENDOR": {
         "Passed": true,
         "actual": "SolidFire",
         "comparator": "==",
         "expected": "SolidFire"
        },
        "BIOS_VERSION": {
         "Passed": true,
         "actual": "1.1.2",
         "comparator": ">=",
         "expected": "1.1.2"
        },
        "BMC_FIRMWARE_REVISION": {
         "Passed": true,
         "actual": "1.6",
         "comparator": ">=",
         "expected": "1.6"
        },
        "BMC_IPMI_VERSION": {
         "Passed": true,
         "actual": "2.0",
         "comparator": ">=",
         "expected": "2.0"
        },
        "CHASSIS_TYPE": {
         "Passed": true,
         "actual": "R620",
         "comparator": "==",
         "expected": "R620"
        },
        "CPU_CORES_00": {
         "Passed": true,
         "actual": "6",
```

```
        "comparator": "==",
        "expected": "6"
       },
       "CPU_CORES_01": {
        "Passed": true,
        "actual": "6",
        "comparator": "==",
        "expected": "6"
       },
       "CPU_CORES_ENABLED_00": {
        "Passed": true,
        "actual": "6",
        "comparator": "==",
        "expected": "6"
       },
       "CPU_CORES_ENABLED_01": {
        "Passed": true,
        "actual": "6",
        "comparator": "==",
        "expected": "6"
       },
       "CPU_MODEL_00": {
        "Passed": true,
        "actual": "Intel(R) Xeon(R) CPU E5-2640 0 @ 2.50GHz",
        "comparator": "==",
        "expected": "Intel(R) Xeon(R) CPU E5-2640 0 @ 2.50GHz"
       },
       "CPU_MODEL_01": {
        "Passed": true,
        "actual": "Intel(R) Xeon(R) CPU E5-2640 0 @ 2.50GHz",
        "comparator": "==",
        "expected": "Intel(R) Xeon(R) CPU E5-2640 0 @ 2.50GHz"
       },
       "CPU_THREADS_00": {
        "Passed": true,
        "actual": "12",
        "comparator": "==",
        "expected": "12"
       },
       "CPU_THREADS_01": {
        "Passed": true,
        "actual": "12",
        "comparator": "==",
        "expected": "12"
       },
       "DRIVE_SIZE_BYTES_SDIMM0": {
```

```
        "Passed": true,
        "actual": "100030242816",
        "comparator": ">=",
        "expected": "100030242816"
       },
       "FIBRE_CHANNEL_FIRMWARE_REVISION": {
        "Passed": true,
        "actual": "FW:v7.04.00",
        "comparator": "==",
        "expected": "FW:v7.04.00"
       },
       "FIBRE_CHANNEL_MODEL": {
        "Passed": true,
        "actual": "QLE2672",
        "comparator": "==",
        "expected": "QLE2672"
       },
       "IDRAC_VERSION": {
        "Passed": true,
        "actual": "1.06.06",
        "comparator": ">=",
        "expected": "1.06.06"
       },
       "LIFECYCLE_VERSION": {
        "Passed": true,
        "actual": "1.0.0.5747",
        "comparator": ">=",
        "expected": "1.0.0.5747"
       },
       "MEMORY_GB": {
        "Passed": true,
        "actual": "32",
        "comparator": ">=",
        "expected": "32"
       },
       "MEMORY_MHZ_00": {
        "Passed": true,
        "actual": "1333",
        "comparator": ">=",
        "expected": "1333"
       },
       "MEMORY_MHZ_01": {
        "Passed": true,
        "actual": "1333",
        "comparator": ">=",
        "expected": "1333"
```

```
        },
        "MEMORY_MHZ_02": {
        "Passed": true,
        "actual": "1333",
        "comparator": ">=",
        "expected": "1333"
       },
       "MEMORY_MHZ_03": {
        "Passed": true,
        "actual": "1333",
        "comparator": ">=",
        "expected": "1333"
       },
       "NETWORK_DRIVER_ETH0": {
        "Passed": true,
        "actual": "bnx2x",
        "comparator": "=~",
        "expected": "^bnx2x$"
       },
       {
        "NETWORK_DRIVER_ETH1":, {...
       },
        "NETWORK_DRIVER_ETH2":, {...
       },
        "NETWORK_DRIVER_ETH3":, {...
       },
        "NETWORK_DRIVER_ETH4":, {...
       },
        "NETWORK_DRIVER_ETH5":, {...
       },
        "NODE_TYPE": {
        "Passed": true,
        "actual": "FC0025",
        "comparator": "==",
        "expected": "FC0025"
       },
       "NUM_CPU": {
        "Passed": true,
        "actual": "2",
        "comparator": "==",
        "expected": "2"
       },
       "NUM_DRIVES": {
        "Passed": true,
        "actual": "0",
        "comparator": "==",
```

```
        "expected": "0"
       },
       "NUM_DRIVES_INTERNAL": {
        "Passed": true,
        "actual": "1",
        "comparator": "==",
        "expected": "1"
       },
       "NUM_FIBRE_CHANNEL_PORTS": {
        "Passed": true,
        "actual": "4",
        "comparator": "==",
        "expected": "4"
       },
       "NVRAM_VENDOR": {
        "Passed": true,
        "actual": "",
        "comparator": "==",
        "expected": ""
       },
       "ROOT_DRIVE_REMOVABLE": {
        "Passed": true,
        "actual": "false",
        "comparator": "==",
        "expected": "false"
       }
       },
       "memory": {
         "firmware_": {
          "capacity": "8323072",
          "date": "03/08/2012",
          "description": "BIOS",
          "physid": "0",
          "size": "65536",
          "vendor": "SolidFire",
          "version": "1.1.2"
       },
       "memory_DMI:1000": {
         "description": "System Memory",
         "physid": "1000",
         "size": "34359738368",
         "slot": "System board or motherboard"
       }
       },
       "network": {
        "network:0_PCI:0000:01:00.0": {
```

```
        "businfo": "pci@0000:01:00.0",
        "capacity": "1000000000",
        "clock": "33000000",
        "description": "Ethernet interface",
        "logicalname": "eth0",
        "physid": "0",
        "product": "NetXtreme II BCM57800 1/10 Gigabit Ethernet",
        "serial": "c8:1f:66:e0:97:2a",
        "vendor": "Broadcom Corporation",
        "version": "10",
        "width": "64"
      },
       "network:0_PCI:0000:41:00.0": {...
      },
       "network:1_PCI:0000:01:00.1": {...
      },
       "network:1_PCI:0000:41:00.1": {...
      },
       "network:2_PCI:0000:01:00.2": {...
      },
       "network:3_PCI:0000:01:00.3": {...
      }
      },
      "networkInterfaces": {
       "Bond10G": {
        "isConfigured": true,
        "isUp": true
      },
      "Bond1G": {
       "isConfigured": true,
       "isUp": true
      },
      "eth0": {
       "isConfigured": true,
       "isUp": true
      },
      "eth1": {...
      },
      "eth2": {...
      },
      "eth3": {...
      },
      "eth4": {...
      },
      "eth5": {...
      }
```

```
      },
      "nvram": {
        "errors": {
         "numOfErrorLogEntries": "0"
        },
        "extended": {
         "dialogVersion": "4",
         "event": [
         {
          "name": "flushToFlash",
          "time": "2015-08-06 01:19:39",
          "value": "0"
         },
         {
          "name": "flushToFlash",
          "time": "2015-08-06 01:26:44",
          "value": "0"
         },
         {... next "flushToFlash"
         },
         {... next "flushToFlash"
         },
         {... next "flushToFlash"
         },
         {... next "flushToFlash"
         },
         {... next "flushToFlash"
         },
         {... next "flushToFlash"
         },
         {... next "flushToFlash"
         }
        ],
        "eventOccurrences": [
         {
           "count": "740",
           "name": "flushToFlash"
         },
         {
           "count": "1",
           "name": "excessiveCurrent"
         }
        ],
        "initialCapacitance": "6.630 F",
        "initialEsr": "0.101 Ohm",
        "measurement": [
```

```
        {
          "level_0": " 0",
          "level_1": " 3969",
          "level_2": " 4631",
          "level_3": " 12875097",
          "level_4": " 1789948",
          "level_5": " 0",
          "level_6": " 0",
          "level_7": " 0",
          "level_8": " 0",
          "level_9": " 0",
          "name": "enterpriseFlashControllerTemperature",
          "recent": "66 C"
      },
      {
          "level_0": " 0",
          "level_1": " 58",
          "level_2": " 1479058",
          "level_3": " 12885356",
          "level_4": " 308293",
          "level_5": " 851",
          "level_6": " 29",
          "level_7": " 0",
          "level_8": " 0",
          "level_9": " 0",
          "name": "capacitor1And2Temperature",
          "recent": "30.69 C"
      },
      {...next temp measurement
      },
      {...next temp measurement
      },
      {...next temp measurement
      },
      }
       "name": "voltageOfCapacitor1",
       "recent": "2.198 V"
      },
      {
       "name": "voltageOfCapacitor2",
       "recent": "2.181 V"
      },
      {
       "name": "voltageOfCapacitor3",
       "recent": "2.189 V"
      },
```

```
      {
       "name": "voltageOfCapacitor4",
       "recent": "2.195 V"
      },
      {
       "level_0": " 4442034",
       "level_1": " 6800018",
       "level_2": " 2846869",
       "level_3": " 119140",
       "level_4": " 29506",
       "level_5": " 428935",
       "level_6": " 7143",
       "level_7": " 0",
       "level_8": " 0",
       "level_9": " 0",
       "name": "capacitorPackVoltage",
       "recent": "8.763 V"
      },
      {
       "level_0": " 0",
       "level_1": " 0",
       "level_2": " 0",
       "level_3": " 0",
       "level_4": " 189",
       "level_5": " 17",
       "level_6": " 36",
       "level_7": " 0",
       "level_8": " 2",
       "level_9": " 490",
       "name": "capacitorPackVoltageAtEndOfFlushToFlash",
       "recent": "4.636 V"
      },
      {
       "name": "currentDerivedFromV3V4",
       "recent": "-0.004 A"
      },
      {
       "level_0": " 230",
       "level_1": " 482",
       "level_2": " 22",
       "level_3": " 0",
       "level_4": " 0",
       "level_5": " 0",
       "level_6": " 0",
       "level_7": " 0",
       "level_8": " 0",
```

```
        "level_9": " 0",
        "name": "derivedEnergy",
        "recent": "172 Joules"
      },
      {...next voltage measurement
      },
      {...next voltage measurement
      },
      {...next voltage measurement
      },
    ],
    "smartCounters": [
      {
        "name": "numberOf512ByteBlocksReadFromDdr",
        "value": "10530088847"
      },
      {
        "name": "numberOf512ByteBlocksWrittenToDdr",
        "value": "1752499453837"
      },
      {
        "name": "numberOfHostReadCommands",
        "value": "235317769"
      },
      {...next smartCounters measurement
      },
      {...next smartCounters measurement
      },
      {...next smartCounters measurement
      },
    ],
    "snapshotTime": "2015-08-20 16:30:01"
  },
  "firmware": {
    "activeSlotNumber": "2",
    "slot1Version": "1e5817bc",
    "slot2Version": "5fb7565c",
    "slot3Version": "1e5817bc",
    "slot4Version": "1e5817bc"
  },
  "identify": {
    "firmwareVersion": "5fb7565c on slot 2",
    "hardwareRevision": "B04",
    "modelNumber": "RMS-200",
    "serialNumber": "0000862"
  },
```

```
    "smart": {
      "availableSpace": "0%",
      "availableSpaceThreshold": "0%",
      "controllerBusyTimeMinutes": "6793",
      "criticalErrorVector": "0x0",
      "mediaErrors": "0",
      "numberOf512ByteBlocksRead": "10530088847",
      "numberOf512ByteBlocksWritten": "1752499439063",
      "numberOfErrorInfoLogs": "1",
      "numberOfHostReadCommands": "235317769",
      "numberOfHostWriteCommands": "126030374065",
      "numberOfPowerCycles": "709",
      "powerOnHours": "11223",
      "temperature": "324 Kelvin",
      "unsafeShutdowns": "357"
       }
       },
       "origin": null,
       "platform": {
        "chassisType": "R620",
        "cpuModel": "Intel(R) Xeon(R) CPU E5-2640 0 @ 2.50GHz",
        "nodeMemoryGB": 32,
        "nodeType": "FC0025"
       },
       "powerSupplies": {
        "PS1 status": {
         "powerSupplyFailureDetected": false,
         "powerSupplyHasAC": true,
         "powerSupplyPredictiveFailureDetected": false,
         "powerSupplyPresent": true,
         "powerSupplyPresentLastCheck": true
       },
        "PS2 status": {
         "powerSupplyFailureDetected": false,
         "powerSupplyHasAC": true,
         "powerSupplyPredictiveFailureDetected": false,
         "powerSupplyPresent": true,
         "powerSupplyPresentLastCheck": true
       }
       },
       "storage": {
        "storage_PCI:0000:00:1f.2": {
         "businfo": "pci@0000:00:1f.2",
         "clock": "66000000",
         "description": "SATA controller",
         "physid": "1f.2",
```

```
        "product": "C600/X79 series chipset 6-Port SATA AHCI Controller",
        "vendor": "Intel Corporation",
        "version": "05",
        "width": "32"
      }
      },
      "system": {
       "ubuntu_DMI:0100": {
        "description": "Rack Mount Chassis",
        "product": "SFx010 ()",
        "serial": "HTW1DZ1",
        "vendor": "SolidFire",
        "width": "64"
      }
      },
      "temperatures": {
       "Exhaust Temp": {
        "baseUnit": "C",
        "threshold": 70,
        "value": 41
      },
      "Inlet Temp": {
       "baseUnit": "C",
       "threshold": 42,
       "value": 18
      }
      },
      "uuid": "4C4C4544-0054-5710-8031-C8C04F445A31"
    },
  "2": {...},                    Storage Node "2"
  "3": {...},                    Storage Node "3"
  "4": {...},                    Storage Node "4"
  "5": {                      Fibre Channel Node
       }
      }
    }
  }
```


GetLldpInfo

`GetLldpInfo` 方法返回类似于以下示例的响应。


```
{
  "id": null,
  "result": {
```

```
    "lldpInfo": {
        "lldpChassis": {
        "local-chassis": [
        {
          "chassis": [
            {
              "capability": [
                {
                  "enabled": false,
                  "type": "Bridge"
                },
                {
                  "enabled": false,
                  "type": "Router"
                },
                {
                  "enabled": false,
                  "type": "Wlan"
                },
                {
                  "enabled": true,
                  "type": "Station"
                }
              ],
              "descr": [
                {
                  "value": "Element OS 11.0"
                }
              ],
              "id": [
                {
                  "type": "mac",
                  "value": "08:00:27:3c:0a:f4"
                }
              ],
              "mgmt-ip": [
                {
                  "value": "10.0.2.15"
                },
                {
                  "value": "fe80::a00:27ff:fe3c:af4"
                }
              ],
              "name": [
                {
                  "value": "SF-93FF"
```

```
                }
              ]
            }
          ],
          "lldp-med": [
            {
              "capability": [
                {
                  "available": true,
                  "type": "Capabilities"
                },
                {
                  "available": true,
                  "type": "Policy"
                },
                {
                  "available": true,
                  "type": "Location"
                },
                {
                  "available": true,
                  "type": "MDI/PSE"
                },
                {
                  "available": true,
                  "type": "MDI/PD"
                },
                {
                  "available": true,
                  "type": "Inventory"
                }
              ],
              "device-type": [
                {
                  "value": "Generic Endpoint (Class I)"
                }
              ],
              "inventory": [
                {
                  "firmware": [
                    {
                      "value": "VirtualBox"
                    }
                  ],
                  "hardware": [
                    {
```

```
                    "value": "1.2"
                  }
                ],
                "manufacturer": [
                  {
                    "value": "innotek GmbH"
                  }
                ],
                "model": [
                  {
                    "value": "VirtualBox"
                  }
                ],
                "serial": [
                  {
                    "value": "0"
                  }
                ],
                "software": [
                  {
                    "value": "4.14.27-solidfire2"
                  }
                ]
              }
            ]
          }
        ]
      }
    ]
  },
  "lldpInterfaces": {
    "lldp": [
      {
        "interface": [
          {
            "age": "0 day, 00:01:04",
            "chassis": [
              {
                "capability": [
                  {
                    "enabled": false,
                    "type": "Bridge"
                  },
                  {
                    "enabled": false,
                    "type": "Router"
```

```
                },
                {
                  "enabled": false,
                  "type": "Wlan"
                },
                {
                  "enabled": true,
                  "type": "Station"
                }
              ],
              "descr": [
                {
                  "value": "Element OS 11.0"
                }
              ],
              "id": [
                {
                  "type": "mac",
                  "value": "08:00:27:3c:0a:f4"
                }
              ],
              "mgmt-ip": [
                {
                  "value": "10.0.2.15"
                },
                {
                  "value": "fe80::a00:27ff:fe3c:af4"
                }
              ],
              "name": [
                {
                  "value": "SF-93FF"
                }
              ]
            }
          ],
          "lldp-med": [
            {
              "capability": [
                {
                  "available": true,
                  "type": "Capabilities"
                },
                {
                  "available": true,
                  "type": "Policy"
```

```
      },
      {
        "available": true,
        "type": "Location"
      },
      {
        "available": true,
        "type": "MDI/PSE"
      },
      {
        "available": true,
        "type": "MDI/PD"
      },
      {
        "available": true,
        "type": "Inventory"
      }
    ],
    "device-type": [
      {
        "value": "Generic Endpoint (Class I)"
      }
    ],
    "inventory": [
      {
        "firmware": [
          {
            "value": "VirtualBox"
          }
        ],
        "hardware": [
          {
            "value": "1.2"
          }
        ],
        "manufacturer": [
          {
            "value": "innotek GmbH"
          }
        ],
        "model": [
          {
            "value": "VirtualBox"
          }
        ],
        "serial": [
```

```
                    {
                      "value": "0"
                    }
                  ],
                  "software": [
                    {
                      "value": "4.14.27-solidfire2"
                    }
                  ]
                }
              ]
            }
          ],
          "name": "eth0",
          "port": [
            {
              "aggregation": [
                {
                  "value": "7"
                }
              ],
              "auto-negotiation": [
                {
                  "advertised": [
                    {
                      "fd": true,
                      "hd": true,
                      "type": "10Base-T"
                    },
                    {
                      "fd": true,
                      "hd": true,
                      "type": "100Base-TX"
                    },
                    {
                      "fd": true,
                      "hd": false,
                      "type": "1000Base-T"
                    }
                  ],
                  "current": [
                    {
                      "value": "full duplex mode"
                    }
                  ],
                  "enabled": true,
```

```
                "supported": true
              }
            ],
            "descr": [
              {
                "value": "eth0"
              }
            ],
            "id": [
              {
                "type": "mac",
                "value": "08:00:27:3c:0a:f4"
              }
            ]
          }
        ],
        "ttl": [
          {
            "ttl": "120"
          }
        ],
        "via": "unknown"
      },
      {
        "age": "17722 days, 17:14:28",
        "chassis": [
          {
            "capability": [
              {
                "enabled": false,
                "type": "Bridge"
              },
              {
                "enabled": false,
                "type": "Router"
              },
              {
                "enabled": false,
                "type": "Wlan"
              },
              {
                "enabled": true,
                "type": "Station"
              }
            ],
            "descr": [
```

```
                {
                  "value": "Element OS 11.0"
                }
              ],
              "id": [
                {
                  "type": "mac",
                  "value": "08:00:27:3c:0a:f4"
                }
              ],
              "mgmt-ip": [
                {
                  "value": "10.0.2.15"
                },
                {
                  "value": "fe80::a00:27ff:fe3c:af4"
                }
              ],
              "name": [
                {
                  "value": "SF-93FF"
                }
              ]
            }
          ],
          "lldp-med": [
            {
              "capability": [
                {
                  "available": true,
                  "type": "Capabilities"
                },
                {
                  "available": true,
                  "type": "Policy"
                },
                {
                  "available": true,
                  "type": "Location"
                },
                {
                  "available": true,
                  "type": "MDI/PSE"
                },
                {
                  "available": true,
```

```
            "type": "MDI/PD"
          },
          {
            "available": true,
            "type": "Inventory"
          }
        ],
        "device-type": [
          {
            "value": "Generic Endpoint (Class I)"
          }
        ],
        "inventory": [
          {
            "firmware": [
              {
                "value": "VirtualBox"
              }
            ],
            "hardware": [
              {
                "value": "1.2"
              }
            ],
            "manufacturer": [
              {
                "value": "innotek GmbH"
              }
            ],
            "model": [
              {
                "value": "VirtualBox"
              }
            ],
            "serial": [
              {
                "value": "0"
              }
            ],
            "software": [
              {
                "value": "4.14.27-solidfire2"
              }
            ]
          }
        ]
```

```
      }
    ],
    "name": "eth1",
    "port": [
      {
        "aggregation": [
          {
            "value": "7"
          }
        ],
        "auto-negotiation": [
          {
            "advertised": [
              {
                "fd": true,
                "hd": true,
                "type": "10Base-T"
              },
              {
                "fd": true,
                "hd": true,
                "type": "100Base-TX"
              },
              {
                "fd": true,
                "hd": false,
                "type": "1000Base-T"
              }
            ],
            "current": [
              {
                "value": "unknown"
              }
            ],
            "enabled": true,
            "supported": true
          }
        ],
        "descr": [
          {
            "value": "eth1"
          }
        ],
        "id": [
          {
            "type": "mac",
```

```
                "value": "08:00:27:36:79:78"
              }
            ]
          }
        ],
        "ttl": [
          {
            "ttl": "120"
          }
        ],
        "via": "unknown"
      },
      {
        "age": "0 day, 00:01:01",
        "chassis": [
          {
            "capability": [
              {
                "enabled": false,
                "type": "Bridge"
              },
              {
                "enabled": false,
                "type": "Router"
              },
              {
                "enabled": false,
                "type": "Wlan"
              },
              {
                "enabled": true,
                "type": "Station"
              }
            ],
            "descr": [
              {
                "value": "Element OS 11.0"
              }
            ],
            "id": [
              {
                "type": "mac",
                "value": "08:00:27:3c:0a:f4"
              }
            ],
            "mgmt-ip": [
```

```
                {
                  "value": "10.0.2.15"
                },
                {
                  "value": "fe80::a00:27ff:fe3c:af4"
                }
              ],
              "name": [
                {
                  "value": "SF-93FF"
                }
              ]
            }
          ],
          "lldp-med": [
            {
              "capability": [
                {
                  "available": true,
                  "type": "Capabilities"
                },
                {
                  "available": true,
                  "type": "Policy"
                },
                {
                  "available": true,
                  "type": "Location"
                },
                {
                  "available": true,
                  "type": "MDI/PSE"
                },
                {
                  "available": true,
                  "type": "MDI/PD"
                },
                {
                  "available": true,
                  "type": "Inventory"
                }
              ],
              "device-type": [
                {
                  "value": "Generic Endpoint (Class I)"
                }
```

```
                ],
                "inventory": [
                  {
                    "firmware": [
                      {
                        "value": "VirtualBox"
                      }
                    ],
                    "hardware": [
                      {
                        "value": "1.2"
                      }
                    ],
                    "manufacturer": [
                      {
                        "value": "innotek GmbH"
                      }
                    ],
                    "model": [
                      {
                        "value": "VirtualBox"
                      }
                    ],
                    "serial": [
                      {
                        "value": "0"
                      }
                    ],
                    "software": [
                      {
                        "value": "4.14.27-solidfire2"
                      }
                    ]
                  }
                ]
              }
            ],
            "name": "eth2",
            "port": [
              {
                "aggregation": [
                  {
                    "value": "6"
                  }
                ],
                "auto-negotiation": [
```

```
                {
                  "advertised": [
                    {
                      "fd": true,
                      "hd": true,
                      "type": "10Base-T"
                    },
                    {
                      "fd": true,
                      "hd": true,
                      "type": "100Base-TX"
                    },
                    {
                      "fd": true,
                      "hd": false,
                      "type": "1000Base-T"
                    }
                  ],
                  "current": [
                    {
                      "value": "full duplex mode"
                    }
                  ],
                  "enabled": true,
                  "supported": true
                }
              ],
              "descr": [
                {
                  "value": "eth2"
                }
              ],
              "id": [
                {
                  "type": "mac",
                  "value": "08:00:27:fc:f0:a9"
                }
              ]
            }
          ],
          "ttl": [
            {
              "ttl": "120"
            }
          ],
          "via": "LLDP"
```

```
            },
            {
              "age": "0 day, 00:01:01",
              "chassis": [
                {
                  "capability": [
                    {
                      "enabled": false,
                      "type": "Bridge"
                    },
                    {
                      "enabled": false,
                      "type": "Router"
                    },
                    {
                      "enabled": false,
                      "type": "Wlan"
                    },
                    {
                      "enabled": true,
                      "type": "Station"
                    }
                  ],
                  "descr": [
                    {
                      "value": "Element OS 11.0"
                    }
                  ],
                  "id": [
                    {
                      "type": "mac",
                      "value": "08:00:27:3c:0a:f4"
                    }
                  ],
                  "mgmt-ip": [
                    {
                      "value": "10.0.2.15"
                    },
                    {
                      "value": "fe80::a00:27ff:fe3c:af4"
                    }
                  ],
                  "name": [
                    {
                      "value": "SF-93FF"
                    }
```

```
                ]
              }
            ],
            "lldp-med": [
              {
                "capability": [
                  {
                    "available": true,
                    "type": "Capabilities"
                  },
                  {
                    "available": true,
                    "type": "Policy"
                  },
                  {
                    "available": true,
                    "type": "Location"
                  },
                  {
                    "available": true,
                    "type": "MDI/PSE"
                  },
                  {
                    "available": true,
                    "type": "MDI/PD"
                  },
                  {
                    "available": true,
                    "type": "Inventory"
                  }
                ],
                "device-type": [
                  {
                    "value": "Generic Endpoint (Class I)"
                  }
                ],
                "inventory": [
                  {
                    "firmware": [
                      {
                        "value": "VirtualBox"
                      }
                    ],
                    "hardware": [
                      {
                        "value": "1.2"
```

```
                }
              ],
              "manufacturer": [
                {
                  "value": "innotek GmbH"
                }
              ],
              "model": [
                {
                  "value": "VirtualBox"
                }
              ],
              "serial": [
                {
                  "value": "0"
                }
              ],
              "software": [
                {
                  "value": "4.14.27-solidfire2"
                }
              ]
            }
          ]
        }
      ],
      "name": "eth3",
      "port": [
        {
          "aggregation": [
            {
              "value": "6"
            }
          ],
          "auto-negotiation": [
            {
              "advertised": [
                {
                  "fd": true,
                  "hd": true,
                  "type": "10Base-T"
                },
                {
                  "fd": true,
                  "hd": true,
                  "type": "100Base-TX"
```

```
                    },
                    {
                      "fd": true,
                      "hd": false,
                      "type": "1000Base-T"
                    }
                  ],
                  "current": [
                    {
                      "value": "full duplex mode"
                    }
                  ],
                  "enabled": true,
                  "supported": true
                }
              ],
              "descr": [
                {
                  "value": "eth3"
                }
              ],
              "id": [
                {
                  "type": "mac",
                  "value": "08:00:27:2c:e4:f8"
                }
              ]
            }
          ],
          "ttl": [
            {
              "ttl": "120"
            }
          ],
          "via": "LLDP"
        }
      ]
    }
  ]
},
"lldpNeighbors": {
  "lldp": [
    {
      "interface": [
        {
          "age": "0 day, 00:04:34",
```

```
            "chassis": [
              {
                "capability": [
                  {
                    "enabled": true,
                    "type": "Bridge"
                  },
                  {
                    "enabled": true,
                    "type": "Router"
                  },
                  {
                    "enabled": true,
                    "type": "Wlan"
                  },
                  {
                    "enabled": false,
                    "type": "Station"
                  }
                ],
                "descr": [
                  {
                    "value": "x86_64"
                  }
                ],
                "id": [
                  {
                    "type": "mac",
                    "value": "50:7b:9d:2b:36:84"
                  }
                ],
                "mgmt-ip": [
                  {
                    "value": "192.168.100.1"
                  },
                  {
                    "value": "fe80::a58e:843:952e:d8eb"
                  }
                ],
                "name": [
                  {
                    "value": "ConventionalWisdom.wlan.netapp.com"
                  }
                ]
              }
            ],
```

```
            "name": "eth2",
            "port": [
              {
                "auto-negotiation": [
                  {
                    "current": [
                      {
                        "value": "full duplex mode"
                      }
                    ],
                    "enabled": false,
                    "supported": false
                  }
                ],
                "descr": [
                  {
                    "value": "vboxnet1"
                  }
                ],
                "id": [
                  {
                    "type": "mac",
                    "value": "0a:00:27:00:00:01"
                  }
                ],
                "ttl": [
                  {
                    "value": "120"
                  }
                ]
              }
            ],
            "rid": "2",
            "via": "LLDP"
          },
          {
            "age": "0 day, 00:01:01",
            "chassis": [
              {
                "capability": [
                  {
                    "enabled": false,
                    "type": "Bridge"
                  },
                  {
                    "enabled": false,
```

```
                "type": "Router"
              },
              {
                "enabled": false,
                "type": "Wlan"
              },
              {
                "enabled": true,
                "type": "Station"
              }
            ],
            "descr": [
              {
                "value": "Element OS 11.0"
              }
            ],
            "id": [
              {
                "type": "mac",
                "value": "08:00:27:3c:0a:f4"
              }
            ],
            "mgmt-ip": [
              {
                "value": "10.0.2.15"
              },
              {
                "value": "fe80::a00:27ff:fe3c:af4"
              }
            ],
            "name": [
              {
                "value": "SF-93FF"
              }
            ]
          }
        ],
        "lldp-med": [
          {
            "capability": [
              {
                "available": true,
                "type": "Capabilities"
              },
              {
                "available": true,
```

```
                "type": "Policy"
              },
              {
                "available": true,
                "type": "Location"
              },
              {
                "available": true,
                "type": "MDI/PSE"
              },
              {
                "available": true,
                "type": "MDI/PD"
              },
              {
                "available": true,
                "type": "Inventory"
              }
            ],
            "device-type": [
              {
                "value": "Generic Endpoint (Class I)"
              }
            ],
            "inventory": [
              {
                "firmware": [
                  {
                    "value": "VirtualBox"
                  }
                ],
                "hardware": [
                  {
                    "value": "1.2"
                  }
                ],
                "manufacturer": [
                  {
                    "value": "innotek GmbH"
                  }
                ],
                "model": [
                  {
                    "value": "VirtualBox"
                  }
                ],
```

```
                "serial": [
                  {
                    "value": "0"
                  }
                ],
                "software": [
                  {
                    "value": "4.14.27-solidfire2"
                  }
                ]
              }
            ]
          }
        ],
        "name": "eth2",
        "port": [
          {
            "aggregation": [
              {
                "value": "6"
              }
            ],
            "auto-negotiation": [
              {
                "advertised": [
                  {
                    "fd": true,
                    "hd": true,
                    "type": "10Base-T"
                  },
                  {
                    "fd": true,
                    "hd": true,
                    "type": "100Base-TX"
                  },
                  {
                    "fd": true,
                    "hd": false,
                    "type": "1000Base-T"
                  }
                ],
                "current": [
                  {
                    "value": "full duplex mode"
                  }
                ],
```

```
                "enabled": true,
                "supported": true
              }
            ],
            "descr": [
              {
                "value": "eth3"
              }
            ],
            "id": [
              {
                "type": "mac",
                "value": "08:00:27:2c:e4:f8"
              }
            ],
            "ttl": [
              {
                "value": "120"
              }
            ]
          }
        ],
        "rid": "1",
        "via": "LLDP"
      },
      {
        "age": "0 day, 00:04:34",
        "chassis": [
          {
            "capability": [
              {
                "enabled": true,
                "type": "Bridge"
              },
              {
                "enabled": true,
                "type": "Router"
              },
              {
                "enabled": true,
                "type": "Wlan"
              },
              {
                "enabled": false,
                "type": "Station"
              }
```

```
                ],
                "descr": [
                  {
                    "value": "x86_64"
                  }
                ],
                "id": [
                  {
                    "type": "mac",
                    "value": "50:7b:9d:2b:36:84"
                  }
                ],
                "mgmt-ip": [
                  {
                    "value": "192.168.100.1"
                  },
                  {
                    "value": "fe80::a58e:843:952e:d8eb"
                  }
                ],
                "name": [
                  {
                    "value": ""
                  }
                ]
              }
            ],
            "name": "eth3",
            "port": [
              {
                "auto-negotiation": [
                  {
                    "current": [
                      {
                        "value": "full duplex mode"
                      }
                    ],
                    "enabled": false,
                    "supported": false
                  }
                ],
                "descr": [
                  {
                    "value": "vboxnet1"
                  }
                ],
```

```
                "id": [
                  {
                    "type": "mac",
                    "value": "0a:00:27:00:00:01"
                  }
                ],
                "ttl": [
                  {
                    "value": "120"
                  }
                ]
              }
            ],
            "rid": "2",
            "via": "LLDP"
          },
          {
            "age": "0 day, 00:01:01",
            "chassis": [
              {
                "capability": [
                  {
                    "enabled": false,
                    "type": "Bridge"
                  },
                  {
                    "enabled": false,
                    "type": "Router"
                  },
                  {
                    "enabled": false,
                    "type": "Wlan"
                  },
                  {
                    "enabled": true,
                    "type": "Station"
                  }
                ],
                "descr": [
                  {
                    "value": "Element OS 11.0"
                  }
                ],
                "id": [
                  {
                    "type": "mac",
```

```
                "value": "08:00:27:3c:0a:f4"
              }
            ],
            "mgmt-ip": [
              {
                "value": "10.0.2.15"
              },
              {
                "value": "fe80::a00:27ff:fe3c:af4"
              }
            ],
            "name": [
              {
                "value": "SF-93FF"
              }
            ]
          }
        ],
        "lldp-med": [
          {
            "capability": [
              {
                "available": true,
                "type": "Capabilities"
              },
              {
                "available": true,
                "type": "Policy"
              },
              {
                "available": true,
                "type": "Location"
              },
              {
                "available": true,
                "type": "MDI/PSE"
              },
              {
                "available": true,
                "type": "MDI/PD"
              },
              {
                "available": true,
                "type": "Inventory"
              }
            ],
```

```
                "device-type": [
                  {
                    "value": "Generic Endpoint (Class I)"
                  }
                ],
                "inventory": [
                  {
                    "firmware": [
                      {
                        "value": "VirtualBox"
                      }
                    ],
                    "hardware": [
                      {
                        "value": "1.2"
                      }
                    ],
                    "manufacturer": [
                      {
                        "value": "innotek GmbH"
                      }
                    ],
                    "model": [
                      {
                        "value": "VirtualBox"
                      }
                    ],
                    "serial": [
                      {
                        "value": "0"
                      }
                    ],
                    "software": [
                      {
                        "value": "4.14.27-solidfire2"
                      }
                    ]
                  }
                ]
              }
            ],
            "name": "eth3",
            "port": [
              {
                "aggregation": [
                  {
```

```
                "value": "6"
              }
            ],
            "auto-negotiation": [
              {
                "advertised": [
                  {
                    "fd": true,
                    "hd": true,
                    "type": "10Base-T"
                  },
                  {
                    "fd": true,
                    "hd": true,
                    "type": "100Base-TX"
                  },
                  {
                    "fd": true,
                    "hd": false,
                    "type": "1000Base-T"
                  }
                ],
                "current": [
                  {
                    "value": "1000BaseTFD"
                  }
                ],
                "enabled": true,
                "supported": true
              }
            ],
            "descr": [
              {
                "value": "eth2"
              }
            ],
            "id": [
              {
                "type": "mac",
                "value": "08:00:27:fc:f0:a9"
              }
            ],
            "ttl": [
              {
                "value": "120"
              }
```

```
                    ]
                  }
                ],
                "rid": "1",
                "via": "LLDP"
              }
            ]
          }
        ]
      }
      }
  }
}
```


GetNetworkConfig

`GetNetworkConfig` 方法返回类似于以下示例的响应。


```
{
    "id": 1,
    "result": {
        "network": {
            "Bond10G": {
                "#default": false,
                "address": "10.1.1.0",
                "auto": true,
                "bond-downdelay": "0",
                "bond-fail_over_mac": "None",
                "bond-miimon": "100",
                "bond-mode": "ActivePassive",
                "bond-primary_reselect": "Failure",
                "bond-slaves": "eth0 eth1",
                "bond-updelay": "200",
                "dns-nameservers": "10.1.1.0, 10.1.1.0",
                "dns-search": "ten.test.company.net., company.net.",
                "family": "inet",
                "gateway": "10.1.1.0",
                "linkSpeed": 10000,
                "macAddress": "c8:1f:66:ee:59:b9",
                "macAddressPermanent": "00:00:00:00:00:00",
                "method": "static",
                "mtu": "9000",
                "netmask": "255.255.240.0",
                "network": "10.1.1.0",
                "physical": {
```

```
                    "address": "10.1.1.0",
                    "macAddress": "c8:1f:66:ee:59:b9",
                    "macAddressPermanent": "00:00:00:00:00:00",
                    "mtu": "9000",
                    "netmask": "255.255.240.0",
                    "network": "10.1.1.0",
                    "upAndRunning": true
                },
                "routes": [],
                "status": "UpAndRunning",
                "symmetricRouteRules": [
                    "ip route add 10.1.1.1/20 dev Bond1G src 10.1.2.2
table Bond1G",
                    "ip rule add from 10.1.1.1 table Bond1G",
                    "ip route add default via 10.1.1.254"
                ],
                "upAndRunning": true,
                "virtualNetworkTag": "0"
            },
            "Bond1G": {
                "#default": true,
                "address": "10.1.1.0",
                "addressV6": "",
                "auto": true,
                "bond-downdelay": "0",
                "bond-fail_over_mac": "None",
                "bond-miimon": "100",
                "bond-mode": "ActivePassive",
                "bond-primary_reselect": "Failure",
                "bond-slaves": "eth2 eth3",
                "bond-updelay": "200",
                "dns-nameservers": "10.1.1.0, 10.1.1.0",
                "dns-search": "ten.test.company.net., company.net.",
                "family": "inet",
                "gateway": "10.1.1.254",
                "gatewayV6": "",
                "linkSpeed": 1000,
                "macAddress": "c8:1f:66:ee:59:bd",
                "macAddressPermanent": "00:00:00:00:00:00",
                "method": "static",
                "mtu": "1500",
                "netmask": "255.255.240.0",
                "network": "10.1.1.0",
                "physical": {
                    "address": "10.1.1.0",
                    "macAddress": "c8:1f:66:ee:59:bd",
```

```
                    "macAddressPermanent": "00:00:00:00:00:00",
                    "mtu": "1500",
                    "netmask": "255.255.240.0",
                    "network": "10.1.1.0",
                    "upAndRunning": true
                },
                "routes": [],
                "status": "UpAndRunning",
                "symmetricRouteRules": [
                    "ip route add 10.1.1.1/20 dev Bond1G src 10.1.2.2
table Bond1G",
                    "ip rule add from 10.1.1.1 table Bond1G",
                    "ip route add default via 10.1.1.254"
                ],
                "upAndRunning": true,
                "virtualNetworkTag": "0"
            },
            "eth0": {
                "auto": true,
                "bond-master": "Bond10G",
                "family": "inet",
                "linkSpeed": 10000,
                "macAddress": "c8:1f:66:ee:59:b9",
                "macAddressPermanent": "c8:1f:66:ee:59:b9",
                "method": "bond",
                "physical": {
                    "address": "0.0.0.0",
                    "macAddress": "c8:1f:66:ee:59:b9",
                    "macAddressPermanent": "c8:1f:66:ee:59:b9",
                    "netmask": "N/A",
                    "network": "N/A",
                    "upAndRunning": true
                },
                "status": "UpAndRunning",
                "upAndRunning": true
            },
            "eth1": {
                "auto": true,
                "bond-master": "Bond10G",
                "family": "inet",
                "linkSpeed": 10000,
                "macAddress": "c8:1f:66:ee:59:b9",
                "macAddressPermanent": "c8:1f:66:ee:59:bb",
                "method": "bond",
                "physical": {
                    "address": "0.0.0.0",
```

```
                "macAddress": "c8:1f:66:ee:59:b9",
                "macAddressPermanent": "c8:1f:66:ee:59:bb",
                "netmask": "N/A",
                "network": "N/A",
                "upAndRunning": true
            },
            "status": "UpAndRunning",
            "upAndRunning": true
        },
        "eth2": {
            "auto": true,
            "bond-master": "Bond1G",
            "family": "inet",
            "linkSpeed": 1000,
            "macAddress": "c8:1f:66:ee:59:bd",
            "macAddressPermanent": "c8:1f:66:ee:59:bd",
            "method": "bond",
            "physical": {
                "address": "0.0.0.0",
                "macAddress": "c8:1f:66:ee:59:bd",
                "macAddressPermanent": "c8:1f:66:ee:59:bd",
                "netmask": "N/A",
                "network": "N/A",
                "upAndRunning": true
            },
            "status": "UpAndRunning",
            "upAndRunning": true
        },
        "eth3": {
            "auto": true,
            "bond-master": "Bond1G",
            "family": "inet",
            "linkSpeed": 1000,
            "macAddress": "c8:1f:66:ee:59:bd",
            "macAddressPermanent": "c8:1f:66:ee:59:bf",
            "method": "bond",
            "physical": {
                "address": "0.0.0.0",
                "macAddress": "c8:1f:66:ee:59:bd",
                "macAddressPermanent": "c8:1f:66:ee:59:bf",
                "netmask": "N/A",
                "network": "N/A",
                "upAndRunning": true
            },
            "status": "UpAndRunning",
            "upAndRunning": true
```

```
            },
            "lo": {
                "auto": true,
                "family": "inet",
                "linkSpeed": 0,
                "macAddress": "00:00:00:00:00:00",
                "macAddressPermanent": "00:00:00:00:00:00",
                "method": "loopback",
                "physical": {
                    "address": "0.0.0.0",
                    "macAddress": "00:00:00:00:00:00",
                    "macAddressPermanent": "00:00:00:00:00:00",
                    "netmask": "N/A",
                    "network": "N/A",
                    "upAndRunning": true
                },
                "status": "UpAndRunning",
                "upAndRunning": true
            }
        }
    }
}
```


GetNodeHardwareInfo （iSCSI 的输出）

iSCSI 的 `GetNodeHardwareInfo` 方法返回类似于以下示例的响应。


```
{
    "id": 1,
    "result": {
        "nodeHardwareInfo": {
            "bus": {
                "core_DMI:0200": {
                    "description": "Motherboard",
                    "physid": "0",
                    "product": "0H47HH",
                    "serial": "..CN7475141I0271.",
                    "vendor": "SolidFire",
                    "version": "A07"
                }
            },
            "driveHardware": [
                {
                "canonicalName": "sda",
```

```
                "connected": true,
                "dev": 2048,
                "devPath": "/dev/slot0",
                "driveEncryptionCapability": "fips",
                "driveType": "Slice",
                "lifeRemainingPercent": 98,
                "lifetimeReadBytes": 0,
                "lifetimeWriteBytes": 14012129342144,
                "name": "scsi-SATA_SAMSUNG_MZ7GE24S1M9NWAG501251",
                "path": "/dev/sda",
                "pathLink": "/dev/slot0",
                "powerOnHours": 15489,
                "product": "SAMSUNG MZ7GE240HMGR-00003",
                "reallocatedSectors": 0,
                "reserveCapacityPercent": 100,
                "scsiCompatId": "scsi-SATA_SAMSUNG_MZ7GE24S1M9NWAG501251",
                "scsiState": "Running",
                "securityAtMaximum": false,
                "securityEnabled": true,
                "securityFrozen": false,
                "securityLocked": false,
                "securitySupported": true,
                "serial": "S1M9NWAG501251",
                "size": 240057409536,
                "slot": 0,
                "uncorrectableErrors": 0,
                "uuid": "789aa05d-e49b-ff4f-f821-f60eed8e43bd",
                "vendor": "Samsung",
                "version": "EXT1303Q"
                },
                {
                "canonicalName": "sda",
                "connected": true,
                "dev": 2048,
                "devPath": "/dev/slot1",
                "driveEncryptionCapability": "fips",
                "driveType": "Slice",
                "lifeRemainingPercent": 98,
                "lifetimeReadBytes": 0,
                "lifetimeWriteBytes": 14112129567184,
                "name": "scsi-SATA_SAMSUNG_MZ7GE24S1M9NWAG501251",
                "path": "/dev/sda",
                "pathLink": "/dev/slot0",
                "powerOnHours": 15489,
                "product": "SAMSUNG MZ7GE240HMGR-00003",
                "reallocatedSectors": 0,
```

```
                "reserveCapacityPercent": 100,
                "scsiCompatId": "scsi-SATA_SAMSUNG_MZ7GE24S1M9NWAG501251",
                "scsiState": "Running",
                "securityAtMaximum": false,
                "securityEnabled": true,
                "securityFrozen": false,
                "securityLocked": false,
                "securitySupported": true,
                "serial": "S1M9NWAG501252",
                "size": 240057409536,
                "slot": 0,
                "uncorrectableErrors": 0,
                "uuid": "789aa05d-e49b-ff4f-f821-f60eed8e43bd",
                "vendor": "Samsung",
                "version": "EXT1303Q"
                }
    }
```


GetNodeHardwareInfo （光纤通道节点的输出）

光纤通道节点的 `GetNodeHardwareInfo` 方法返回类似于以下示例的响应。


```
{
"id": null,
"result": {
"nodeHardwareInfo": {
"bus": {
"core_DMI:0200": {
"description": "Motherboard",
"physid": "0",
"product": "0H47HH",
"serial": "..CN747513AA0541.",
"version": "A07"
},
"fiber:0_PCI:0000:04:00.0": {
"businfo": "pci@0000:04:00.0",
"clock": "33000000",
"description": "Fibre Channel",
"physid": "0",
"product": "ISP8324-based 16Gb Fibre Channel to PCI Express Adapter",
"vendor": "QLogic Corp.",
"version": "02",
"width": "64"
},
```

```
"fiber:0_PCI:0000:42:00.0": {
"businfo": "pci@0000:42:00.0",
"clock": "33000000",
"description": "Fibre Channel",
"physid": "0",
"product": "ISP8324-based 16Gb Fibre Channel to PCI Express Adapter",
"vendor": "QLogic Corp.",
"version": "02",
"width": "64"
},
"fiber:1_PCI:0000:04:00.1": {
"businfo": "pci@0000:04:00.1",
"clock": "33000000",
"description": "Fibre Channel",
"physid": "0.1",
"product": "ISP8324-based 16Gb Fibre Channel to PCI Express Adapter",
"vendor": "QLogic Corp.",
"version": "02",
"width": "64"
},
"fiber:1_PCI:0000:42:00.1": {
"businfo": "pci@0000:42:00.1",
"clock": "33000000",
"description": "Fibre Channel",
"physid": "0.1",
"product": "ISP8324-based 16Gb Fibre Channel to PCI Express Adapter",
"vendor": "QLogic Corp.",
"version": "02",
"width": "64"
}
},
"fans": {
"Fan1A RPM": {
"baseUnit": "RPM",
"threshold": 840,
"value": 3360
},
"Fan1B RPM": {
"baseUnit": "RPM",
"threshold": 840,
"value": 3120
}
},
"fibreChannelPorts": [
{
"firmware": "7.04.00 (d0d5)",
```

```
"hbaPort": 1,
"internalPortID": 2,
"model": "QLE2672",
"nPortID": "0x060019",
"nodeID": 6,
"pciSlot": 3,
"serial": "BFE1335E04217",
"speed": "8 Gbit",
"state": "Online",
"switchWwn": "20:01:00:2a:6a:9c:71:01",
"wwnn": "5f:47:ac:c8:30:26:c9:00",
"wwpn": "5f:47:ac:c0:30:26:c9:0a"
},
{
"firmware": "7.04.00 (d0d5)",
"hbaPort": 2,
"internalPortID": 3,
"model": "QLE2672",
"nPortID": "0xc70019",
"nodeID": 6,
"pciSlot": 3,
"serial": "BFE1335E04217",
"speed": "8 Gbit",
"state": "Online",
"switchWwn": "20:01:00:2a:6a:98:a3:41",
"wwnn": "5f:47:ac:c8:30:26:c9:00",
"wwpn": "5f:47:ac:c0:30:26:c9:0b"
},
{
"firmware": "7.04.00 (d0d5)",
"hbaPort": 1,
"internalPortID": 0,
"model": "QLE2672",
"nPortID": "0xc70017",
"nodeID": 6,
"pciSlot": 2,
"serial": "BFE1341E09515",
"speed": "8 Gbit",
"state": "Online",
"switchWwn": "20:01:00:2a:6a:98:a3:41",
"wwnn": "5f:47:ac:c8:30:26:c9:00",
"wwpn": "5f:47:ac:c0:30:26:c9:08"
},
{
"firmware": "7.04.00 (d0d5)",
"hbaPort": 2,
```

```
"internalPortID": 1,
"model": "QLE2672",
"nPortID": "0x060017",
"nodeID": 6,
"pciSlot": 2,
"serial": "BFE1341E09515",
"speed": "8 Gbit",
"state": "Online",
"switchWwn": "20:01:00:2a:6a:9c:71:01",
"wwnn": "5f:47:ac:c8:30:26:c9:00",
"wwpn": "5f:47:ac:c0:30:26:c9:09"
}
],
"memory": {
"firmware_": {
"capacity": "8323072",
"date": "08/29/2013",
"description": "BIOS",
"physid": "0",
"size": "65536",
"version": "2.0.19"
},
"memory_DMI:1000": {
"description": "System Memory",
"physid": "1000",
"size": "34359738368",
"slot": "System board or motherboard"
}
},
"network": {
"network:0_": {
"description": "Ethernet interface",
"logicalname": "Bond1G",
"physid": "1",
"serial": "c8:1f:66:df:04:da"
},
"network:0_PCI:0000:01:00.0": {
"businfo": "pci@0000:01:00.0",
"capacity": "1000000000",
"clock": "33000000",
"description": "Ethernet interface",
"logicalname": "eth0",
"physid": "0",
"product": "NetXtreme II BCM57800 1/10 Gigabit Ethernet",
"serial": "c8:1f:66:df:04:d6",
"vendor": "Broadcom Corporation",
```

```
"version": "10",
"width": "64"
},
"network:0_PCI:0000:41:00.0": {
"businfo": "pci@0000:41:00.0",
"capacity": "1000000000",
"clock": "33000000",
"description": "Ethernet interface",
"logicalname": "eth4",
"physid": "0",
"product": "NetXtreme II BCM57810 10 Gigabit Ethernet",
"serial": "00:0a:f7:41:7a:30",
"vendor": "Broadcom Corporation",
"version": "10",
"width": "64"
},
"network:1_": {
"description": "Ethernet interface",
"logicalname": "Bond10G",
"physid": "2",
"serial": "c8:1f:66:df:04:d6"
},
"network:1_PCI:0000:01:00.1": {
"businfo": "pci@0000:01:00.1",
"capacity": "1000000000",
"clock": "33000000",
"description": "Ethernet interface",
"logicalname": "eth1",
"physid": "0.1",
"product": "NetXtreme II BCM57800 1/10 Gigabit Ethernet",
"serial": "c8:1f:66:df:04:d8",
"vendor": "Broadcom Corporation",
"version": "10",
"width": "64"
},
"network:1_PCI:0000:41:00.1": {
"businfo": "pci@0000:41:00.1",
"capacity": "1000000000",
"clock": "33000000",
"description": "Ethernet interface",
"logicalname": "eth5",
"physid": "0.1",
"product": "NetXtreme II BCM57810 10 Gigabit Ethernet",
"serial": "00:0a:f7:41:7a:32",
"vendor": "Broadcom Corporation",
"version": "10",
```

```
"width": "64"
},
"network:2_PCI:0000:01:00.2": {
"businfo": "pci@0000:01:00.2",
"capacity": "1000000000",
"clock": "33000000",
"description": "Ethernet interface",
"logicalname": "eth2",
"physid": "0.2",
"product": "NetXtreme II BCM57800 1/10 Gigabit Ethernet",
"serial": "c8:1f:66:df:04:da",
"size": "1000000000",
"vendor": "Broadcom Corporation",
"version": "10",
"width": "64"
},
"network:3_PCI:0000:01:00.3": {
"businfo": "pci@0000:01:00.3",
"capacity": "1000000000",
"clock": "33000000",
"description": "Ethernet interface",
"logicalname": "eth3",
"physid": "0.3",
"product": "NetXtreme II BCM57800 1/10 Gigabit Ethernet",
"serial": "c8:1f:66:df:04:dc",
"size": "1000000000",
"vendor": "Broadcom Corporation",
"version": "10",
"width": "64"
}
},
"networkInterfaces": {
"Bond10G": {
"isConfigured": true,
"isUp": true
},
"Bond1G": {
"isConfigured": true,
"isUp": true
},
"eth0": {
"isConfigured": true,
"isUp": true
},
"eth1": {
"isConfigured": true,
```

```
"isUp": true
},
"eth2": {
"isConfigured": true,
"isUp": true
},
"eth3": {
"isConfigured": true,
"isUp": true
},
"eth4": {
"isConfigured": true,
"isUp": true
},
"eth5": {
"isConfigured": true,
"isUp": true
}
},
"platform": {
"chassisType": "R620",
"cpuModel": "Intel(R) Xeon(R) CPU E5-2640 0 @ 2.50GHz",
"nodeMemoryGB": 32,
"nodeType": "SFFC"
},
"powerSupplies": {
"PS1 status": {
"powerSupplyFailureDetected": false,
"powerSupplyHasAC": true,
"powerSupplyPredictiveFailureDetected": false,
"powerSupplyPresent": true
},
"PS2 status": {
"powerSupplyFailureDetected": false,
"powerSupplyHasAC": true,
"powerSupplyPredictiveFailureDetected": false,
"powerSupplyPresent": true
}
},
"storage": {
"storage_PCI:0000:00:1f.2": {
"businfo": "pci@0000:00:1f.2",
"clock": "66000000",
"description": "SATA controller",
"physid": "1f.2",
"product": "C600/X79 series chipset 6-Port SATA AHCI Controller",
```

```
"vendor": "Intel Corporation",
"version": "05",
"width": "32"
}
},
"system": {
"fcn-2_DMI:0100": {
"description": "Rack Mount Chassis",
"product": "(SKU=NotProvided;ModelName=)",
"serial": "HTX1DZ1",
"width": "64"
}
},
"temperatures": {
"Exhaust Temp": {
"baseUnit": "C",
"threshold": 70,
"value": 38
},
"Inlet Temp": {
"baseUnit": "C",
"threshold": 42,
"value": 13
  },
"uuid": "4C4C4544-004D-5310-8052-C4C04F335431"
    }
   }
  }
 }
}
```


GetNvramInfo

`GetNvramInfo` 方法返回类似于以下示例的响应。


```
{
  id: 1,
  result: {
    nvramInfo: {
      details: {
         errors: {
           numOfErrorLogEntries: "0"
         },
         extended: {
            dialogVersion: "4",
```

```
            event: [
                {
                    name: "flushToFlash",
                    time: "2014-02-24 20:30:28",
                    value: "0"
        },
        {
                    name: "flushToFlash",
                    time: "1946-02-06 17:16:42",
                    value: "0"
        },
        {
                    name: "flushToFlash",
                    time: "2014-02-25 00:48:06",
                    value: "0"
        },
        {
                    name: "flushToFlash",
                    time: "2014-02-25 15:44:07",
                    value: "0"
        },
        {
                    name: "flushToFlash",
                    time: "2014-03-17 17:21:46",
                    value: "0"
        },
        {
                    name: "flushToFlash",
                    time: "2014-03-17 17:59:30",
                    value: "0"
        },
        {
                    name: "flushToFlash",
                    time: "2014-03-17 18:06:27",
                    value: "0"
        },
        {
                    name: "flushToFlash",
                    time: "2014-03-17 21:43:17",
                    value: "0"
        },
        {
                    name: "excessiveCurrent",
                    time: "2014-02-25 00:00:29",
                    value: "39"
        },
```

```
          {
                   name: "excessiveCurrent",
                   time: "2014-03-01 00:00:24",
                   value: "23"
          }
       ],
            eventOccurrences: [
         {
                  count: "15",
                  name: "flushToFlash"
         },
         {
                  count: "2",
                  name: "excessiveCurrent"
         }
                  initialCapacitance: "6.653 F",
                  initialEsr: "0.097 Ohm",
            measurement: [
         {
                  level_0: " 0",
                  level_1: " 112",
                  level_2: " 670919",
                  level_3: " 455356",
                  level_4: " 90215",
                  level_5: " 0",
                  level_6: " 0",
                  level_7: " 0",
                  level_8: " 0",
                  level_9: " 0",
                  name: "enterpriseFlashControllerTemperature",
                  recent: "64 C"
         },
         {
                  level_0: " 0",
                  level_1: " 27",
                  level_2: " 456896",
                  level_3: " 717565",
                  level_4: " 39422",
                  level_5: " 2692",
                  level_6: " 0",
                  level_7: " 0",
                  level_8: " 0",
                  level_9: " 0",
                  name: "capacitor1And2Temperature",
                  recent: "28.64 C"
         },
```

```
        {
                level_0: " 0",
                level_1: " 2080",
                level_2: " 907196",
                level_3: " 280178",
                level_4: " 26539",
                level_5: " 609",
                level_6: " 0",
                level_7: " 0",
                level_8: " 0",
                level_9: " 0",
                name: "capacitor3And4Temperature",
                recent: "28.60 C"
        },
        {
                errorPeriod: {
                    duration: "24",
                    startTime: "2014-02-06 00:23:54",
                    worst: "8"
                },
                level_0: " 0",
                level_1: " 839",
                level_2: " 272794",
                level_3: " 404758",
                level_4: " 35216",
                level_5: " 377818",
                level_6: " 103891",
                level_7: " 21274",
                level_8: " 12",
                level_9: " 0",
                name: "rearVentAmbientTemperature",
                recent: "46.82 C"
         },
         {
                level_0: " 0",
                level_1: " 742749",
                level_2: " 460016",
                level_3: " 13837",
                level_4: " 0",
                level_5: " 0",
                level_6: " 0",
                level_7: " 0",
                level_8: " 0",
                level_9: " 0",
                name: "rms200BoardTemperature",
                recent: "50.62 C"
```

```
        },
        {
                 name: "voltageOfCapacitor1",
                 recent: "2.308 V"
        },
        {
                 name: "voltageOfCapacitor2",
                 recent: "2.305 V"},
        {
                 name: "voltageOfCapacitor3",
                 recent: "2.314 V"
        },
        {
                 name: "voltageOfCapacitor4",
                 recent: "2.307 V"
        },
        {
                level_0: " 175052",
                level_1: " 51173",
                level_2: " 435788",
                level_3: " 12766",
                level_4: " 4",
                level_5: " 6",
                level_6: " 541813",
                level_7: " 0",
                level_8: " 0",
                level_9: " 0",
                name: "capacitorPackVoltage",
                recent: "9.233 V"
         },
         {
                level_0: " 0",
                level_1: " 0",
                level_2: " 0",
                level_3: " 0",
                level_4: " 0",
                level_5: " 0",
                level_6: " 4",
                level_7: " 1",
                level_8: " 4",
                level_9: " 6",
                name: "capacitorPackVoltageAtEndOfFlushToFlash",
                recent: "5.605 V"
        },
        {
                name: "currentDerivedFromV3V4",
```

```
                recent: "0.000 A"
        },
        {
                level_0: " 7",
                level_1: " 4",
                level_2: " 3",
                level_3: " 1",
                level_4: " 0",
                level_5: " 0",
                level_6: " 0",
                level_7: " 0",
                level_8: " 0",
                level_9: " 0",
                name: "derivedEnergy",
                recent: "175 Joules"
        },
        {
                level_0: " 0",
                level_1: " 0",
                level_2: " 0",
                level_3: " 0",
                level_4: " 0",
                level_5: " 0",
                level_6: " 0",
                level_7: " 17",
                level_8: " 19",
                level_9: " 7",
                name: "derivedCapacitanceOfThePack",
                recent: "5.959 F"
         },
         {
                level_0: " 0",
                level_1: " 43",
                level_2: " 0",
                level_3: " 0",
                level_4: " 0",
                level_5: " 0",
                level_6: " 0",
                level_7: " 0",
                level_8: " 0",
                level_9: " 0",
                name: "derivedEsrOfCapacitorPack",
                recent: "0.104 Ohm"
        },
        {
                level_0: " 0",
```

```
                level_1: " 0",
                level_2: " 0",
                level_3: " 0",
                level_4: " 15",
                level_5: " 0",
                level_6: " 0",
                level_7: " 0",
                level_8: " 0",
                level_9: " 0",
                name: "timeToRunFlushToFlash",
                recent: "22.40 Seconds"
        },
        {
                level_0: " 0",
                level_1: " 0",
                level_2: " 7",
                level_3: " 0",
                level_4: " 0",
                level_5: " 0",
                level_6: " 0",
                level_7: " 0",
                level_8: " 0",
                level_9: " 0",
                name: "timeToRunRestore",
                recent: "20.44 Seconds"
        },
        {
                level_0: " 0",
                level_1: " 1",
                level_2: " 3",
                level_3: " 2",
                level_4: " 0",
                level_5: " 0",
                level_6: " 0",
                level_7: " 0",
                level_8: " 0",
                level_9: " 1",
                name: "timeToChargeCapacitors",
                recent: "48 Seconds"
        },
        {
                level_0: " 448586",
                level_1: " 2998",
                level_2: " 0",
                level_3: " 0",
                level_4: " 0",
```

```
                level_5: " 0",
                level_6: " 0",
                level_7: " 0",
                level_8: " 0",
                level_9: " 0",
                name: "correctableBitsInErrorOnReadingAPage"
        },
        {
                level_0: " 2998",
                level_1: " 0",
                level_2: " 0",
                level_3: " 0",
                level_4: " 0",
                level_5: " 0",
                level_6: " 0",
                level_7: " 0",
                level_8: " 0",
                level_9: " 0",
                name:
"correctableBitsInErrorOnReadingTheWorstBchRegionOfAPage"
        },
        {
                level_0: " 0",
                level_1: " 37",
                level_2: " 280274",
                level_3: " 422999",
                level_4: " 245814",
                level_5: " 242470",
                level_6: " 24447",
                level_7: " 561",
                level_8: " 0",
                level_9: " 0",
                name: "fanInletAmbientTemperature",
                recent: "41.74 C"
        }
        ],
                predictedCapacitanceDepletion: "504328 uF",
                smartCounters: [
        {
                name: "numberOf512ByteBlocksReadFromDdr",
                value: "218284648"
        },
        {
                name: "numberOf512ByteBlocksWrittenToDdr",
                value: "12031567354"
        },
```

```
        {
                name: "numberOfHostReadCommands",
                value: "5366315"
        },
        {
                name: "numberOfHostWriteCommands",
                value: "1266099334"
        },
        {
                name: "controllerBusyTimeMinutes",
                value: "0"
        },
        {
                name: "numberOfPowerCycles",
                value: "13"
        },
        {
                name: "powerOnHours",
                value: "1009"
        },
        {
                name: "unsafeShutdowns",
                value: "5"
        },
        {
                name: "mediaErrors",
                value: "0"
        },
        {
                name: "numberOfErrorLogs",
                value: "2"
        }
        ],
         snapshotTime: "2014-03-20 16:43:49"
    },
    firmware: {
         activeSlotNumber: "2",
         slot1Version: "1e5817bc",
         slot2Version: "1e0d70ac",
         slot3Version: "1e5817bc",
         slot4Version: "1e5817bc"
   },
    smart: {
        availableSpace: "0%",
        availableSpaceThreshold: "0%",
        controllerBusyTimeMinutes: "0",
```

```
          criticalErrorVector: "0x0",
          mediaErrors: "0",
          numberOf512ByteBlocksRead: "218284648",
          numberOf512ByteBlocksWritten: "12031567354",
          numberOfErrorInfoLogs: "2",
          numberOfHostReadCommands: "5366315",
          numberOfHostWriteCommands: "1266099334",
          numberOfPowerCycles: "13",
          powerOnHours: "1009",
          temperature: "323 Kelvin",
          unsafeShutdowns: "5"
       }
      },
      status: "Warning",
      statusInfo: {
      warning: [
          "excessiveCurrent (2x)"
        ]
      },
      type: "RMS-200"
     }
   }
 }
```


ListActiveNodes

`ListActiveNodes` 方法返回类似于以下示例的响应。


```
 {
    "id": 1,
    "result": {
        "nodes": [
            {
                "associatedFServiceID": 0,
                "associatedMasterServiceID": 1,
                "attributes": {},
                "cip": "172.27.21.23",
                "cipi": "Bond10G",
                "fibreChannelTargetPortGroup": null,
                "mip": "172.27.1.23",
                "mipi": "Bond1G",
                "name": "PSN-1-23",
                "nodeID": 1,
                "platformInfo": {
                    "chassisType": "R620",
```

```
                    "cpuModel": "Intel(R) Xeon(R) CPU E5-2640 0 @
2.50GHz",
                    "nodeMemoryGB": 72,
                    "nodeType": "SF3010"
                },
                "sip": "172.27.21.23",
                "sipi": "Bond10G",
                "softwareVersion": "9.0.0.1298",
                "uuid": "4C4C4544-0056-3810-804E-B5C04F4C5631",
                "virtualNetworks": [
                    {
                        "address": "10.1.2.4",
                        "virtualNetworkID": 1
                    },
                    {
                        "address": "10.2.2.10",
                        "virtualNetworkID": 2
                    }
                ]
            },
            {
                "associatedFServiceID": 0,
                "associatedMasterServiceID": 4,
                "attributes": {},
                "cip": "172.27.21.24",
                "cipi": "Bond10G",
                "fibreChannelTargetPortGroup": null,
                "mip": "172.27.1.24",
                "mipi": "Bond1G",
                "name": "PSN-1-24",
                "nodeID": 2,
                "platformInfo": {
                    "chassisType": "R620",
                    "cpuModel": "Intel(R) Xeon(R) CPU E5-2640 0 @
2.50GHz",
                    "nodeMemoryGB": 72,
                    "nodeType": "SF3010"
                },
                "sip": "172.27.21.24",
                "sipi": "Bond10G",
                "softwareVersion": "9.0.0.1298",
                "uuid": "4C4C4544-0042-4210-804E-C3C04F4C5631",
                "virtualNetworks": [
                    {
                        "address": "10.1.2.5",
                        "virtualNetworkID": 1
```

```
                    },
                    {
                        "address": "10.2.2.11",
                        "virtualNetworkID": 2
                    }
                ]
            },
            {
                "associatedFServiceID": 0,
                "associatedMasterServiceID": 2,
                "attributes": {},
                "cip": "172.27.21.25",
                "cipi": "Bond10G",
                "fibreChannelTargetPortGroup": null,
                "mip": "172.27.1.25",
                "mipi": "Bond1G",
                "name": "PSN-1-25",
                "nodeID": 3,
                "platformInfo": {
                    "chassisType": "R620",
                    "cpuModel": "Intel(R) Xeon(R) CPU E5-2640 0 @
2.50GHz",
                    "nodeMemoryGB": 72,
                    "nodeType": "SF3010"
                },
                "sip": "172.27.21.25",
                "sipi": "Bond10G",
                "softwareVersion": "9.0.0.1298",
                "uuid": "4C4C4544-0053-4210-8051-C6C04F515631",
                "virtualNetworks": [
                    {
                        "address": "10.1.2.6",
                        "virtualNetworkID": 1
                    },
                    {
                        "address": "10.2.2.12",
                        "virtualNetworkID": 2
                    }
                ]
            },
            {
                "associatedFServiceID": 0,
                "associatedMasterServiceID": 3,
                "attributes": {},
                "cip": "172.27.21.26",
                "cipi": "Bond10G",
```

```
                "fibreChannelTargetPortGroup": null,
                "mip": "172.27.1.26",
                "mipi": "Bond1G",
                "name": "PSN-1-26",
                "nodeID": 4,
                "platformInfo": {
                    "chassisType": "R620",
                    "cpuModel": "Intel(R) Xeon(R) CPU E5-2640 0 @
2.50GHz",
                    "nodeMemoryGB": 72,
                    "nodeType": "SF3010"
                },
                "sip": "172.27.21.26",
                "sipi": "Bond10G",
                "softwareVersion": "9.0.0.1298",
                "uuid": "4C4C4544-0056-3810-804E-B4C04F4C5631",
                "virtualNetworks": [
                    {
                        "address": "10.1.2.7",
                        "virtualNetworkID": 1
                    },
                    {
                        "address": "10.2.2.13",
                        "virtualNetworkID": 2
                    }
                ]
            }
        ]
    }
 }
```


ListActiveVolumes

`ListActiveVolumes` 方法返回类似于以下示例的响应。


```
{
    "id": 1,
    "result": {
        "volumes": [
            {
                "access": "readWrite",
                "accountID": 1,
                "attributes": {},
                "blockSize": 4096,
                "createTime": "2016-06-23T14:19:12Z",
```

```
                "deleteTime": "",
                "enable512e": false,
                "iqn": "iqn.2010-01.com.solidfire:0oto.hulkdemo1.1",
                "name": "HulkDemo1",
                "purgeTime": "",
                "qos": {
                    "burstIOPS": 1500,
                    "burstTime": 60,
                    "curve": {
                        "4096": 100,
                        "8192": 160,
                        "16384": 270,
                        "32768": 500,
                        "65536": 1000,
                        "131072": 1950,
                        "262144": 3900,
                        "524288": 7600,
                        "1048576": 15000
                    },
                    "maxIOPS": 1000,
                    "minIOPS": 100
                },
                "scsiEUIDeviceID": "306f746f00000001f47acc0100000000",
                "scsiNAADeviceID": "6f47acc100000000306f746f00000001",
                "sliceCount": 1,
                "status": "active",
                "totalSize": 53687091200,
                "virtualVolumeID": null,
                "volumeAccessGroups": [
                    1
                ],
                "volumeID": 1,
                "volumePairs": []
            },
            {
                "access": "readWrite",
                "accountID": 1,
                "attributes": {},
                "blockSize": 4096,
                "createTime": "2016-06-23T14:19:14Z",
                "deleteTime": "",
                "enable512e": false,
                "iqn": "iqn.2010-01.com.solidfire:0oto.hulkdemo6.6",
                "name": "HulkDemo6",
                "purgeTime": "",
                "qos": {
```

```
                "burstIOPS": 1500,
                "burstTime": 60,
                "curve": {
                    "4096": 100,
                    "8192": 160,
                    "16384": 270,
                    "32768": 500,
                    "65536": 1000,
                    "131072": 1950,
                    "262144": 3900,
                    "524288": 7600,
                    "1048576": 15000
                },
                "maxIOPS": 1000,
                "minIOPS": 100
            },
            "scsiEUIDeviceID": "306f746f00000006f47acc0100000000",
            "scsiNAADeviceID": "6f47acc100000000306f746f00000006",
            "sliceCount": 1,
            "status": "active",
            "totalSize": 53687091200,
            "virtualVolumeID": null,
            "volumeAccessGroups": [
                1
            ],
            "volumeID": 6,
            "volumePairs": []
        },
        {
            "access": "readWrite",
            "accountID": 1,
            "attributes": {},
            "blockSize": 4096,
            "createTime": "2016-06-23T14:19:14Z",
            "deleteTime": "",
            "enable512e": false,
            "iqn": "iqn.2010-01.com.solidfire:0oto.hulkdemo7.7",
            "name": "HulkDemo7",
            "purgeTime": "",
            "qos": {
                "burstIOPS": 1500,
                "burstTime": 60,
                "curve": {
                    "4096": 100,
                    "8192": 160,
                    "16384": 270,
```

```
                    "32768": 500,
                    "65536": 1000,
                    "131072": 1950,
                    "262144": 3900,
                    "524288": 7600,
                    "1048576": 15000
                },
                "maxIOPS": 1000,
                "minIOPS": 100
            },
            "scsiEUIDeviceID": "306f746f00000007f47acc0100000000",
            "scsiNAADeviceID": "6f47acc100000000306f746f00000007",
            "sliceCount": 1,
            "status": "active",
            "totalSize": 53687091200,
            "virtualVolumeID": null,
            "volumeAccessGroups": [
                1
            ],
            "volumeID": 7,
            "volumePairs": []
        },
        {
            "access": "readWrite",
            "accountID": 1,
            "attributes": {},
            "blockSize": 4096,
            "createTime": "2016-06-23T14:19:15Z",
            "deleteTime": "",
            "enable512e": false,
            "iqn": "iqn.2010-01.com.solidfire:0oto.hulkdemo8.8",
            "name": "HulkDemo8",
            "purgeTime": "",
            "qos": {
                "burstIOPS": 1500,
                "burstTime": 60,
                "curve": {
                    "4096": 100,
                    "8192": 160,
                    "16384": 270,
                    "32768": 500,
                    "65536": 1000,
                    "131072": 1950,
                    "262144": 3900,
                    "524288": 7600,
                    "1048576": 15000
```

```
                    },
                    "maxIOPS": 1000,
                    "minIOPS": 100
                },
                "scsiEUIDeviceID": "306f746f00000008f47acc0100000000",
                "scsiNAADeviceID": "6f47acc100000000306f746f00000008",
                "sliceCount": 1,
                "status": "active",
                "totalSize": 53687091200,
                "virtualVolumeID": null,
                "volumeAccessGroups": [
                    1
                ],
                "volumeID": 8,
                "volumePairs": []
            },
            {
                "access": "readWrite",
                "accountID": 1,
                "attributes": {},
                "blockSize": 4096,
                "createTime": "2016-06-23T14:19:15Z",
                "deleteTime": "",
                "enable512e": false,
                "iqn": "iqn.2010-01.com.solidfire:0oto.hulkdemo9.9",
                "name": "HulkDemo9",
                "purgeTime": "",
                "qos": {
                    "burstIOPS": 1500,
                    "burstTime": 60,
                    "curve": {
                        "4096": 100,
                        "8192": 160,
                        "16384": 270,
                        "32768": 500,
                        "65536": 1000,
                        "131072": 1950,
                        "262144": 3900,
                        "524288": 7600,
                        "1048576": 15000
                    },
                    "maxIOPS": 1000,
                    "minIOPS": 100
                },
                "scsiEUIDeviceID": "306f746f00000009f47acc0100000000",
                "scsiNAADeviceID": "6f47acc100000000306f746f00000009",
```

```
                "sliceCount": 1,
                "status": "active",
                "totalSize": 53687091200,
                "virtualVolumeID": null,
                "volumeAccessGroups": [
                    1
                ],
                "volumeID": 9,
                "volumePairs": []
            },
            {
                "access": "readWrite",
                "accountID": 1,
                "attributes": {},
                "blockSize": 4096,
                "createTime": "2016-06-23T14:19:16Z",
                "deleteTime": "",
                "enable512e": false,
                "iqn": "iqn.2010-01.com.solidfire:0oto.hulkdemo12.12",
                "name": "HulkDemo12",
                "purgeTime": "",
                "qos": {
                    "burstIOPS": 1500,
                    "burstTime": 60,
                    "curve": {
                        "4096": 100,
                        "8192": 160,
                        "16384": 270,
                        "32768": 500,
                        "65536": 1000,
                        "131072": 1950,
                        "262144": 3900,
                        "524288": 7600,
                        "1048576": 15000
                    },
                    "maxIOPS": 1000,
                    "minIOPS": 100
                },
                "scsiEUIDeviceID": "306f746f0000000cf47acc0100000000",
                "scsiNAADeviceID": "6f47acc100000000306f746f0000000c",
                "sliceCount": 1,
                "status": "active",
                "totalSize": 53687091200,
                "virtualVolumeID": null,
                "volumeAccessGroups": [
                    1
```

```
                ],
                "volumeID": 12,
                "volumePairs": []
            },
            {
                "access": "readWrite",
                "accountID": 1,
                "attributes": {},
                "blockSize": 4096,
                "createTime": "2016-06-23T14:19:18Z",
                "deleteTime": "",
                "enable512e": false,
                "iqn": "iqn.2010-01.com.solidfire:0oto.hulkdemo16.16",
                "name": "HulkDemo16",
                "purgeTime": "",
                "qos": {
                    "burstIOPS": 1500,
                    "burstTime": 60,
                    "curve": {
                        "4096": 100,
                        "8192": 160,
                        "16384": 270,
                        "32768": 500,
                        "65536": 1000,
                        "131072": 1950,
                        "262144": 3900,
                        "524288": 7600,
                        "1048576": 15000
                    },
                    "maxIOPS": 1000,
                    "minIOPS": 100
                },
                "scsiEUIDeviceID": "306f746f00000010f47acc0100000000",
                "scsiNAADeviceID": "6f47acc100000000306f746f00000010",
                "sliceCount": 1,
                "status": "active",
                "totalSize": 53687091200,
                "virtualVolumeID": null,
                "volumeAccessGroups": [
                    1
                ],
                "volumeID": 16,
                "volumePairs": []
            },
            {
                "access": "readWrite",
```

```
            "accountID": 1,
            "attributes": {},
            "blockSize": 4096,
            "createTime": "2016-06-23T14:19:18Z",
            "deleteTime": "",
            "enable512e": false,
            "iqn": "iqn.2010-01.com.solidfire:0oto.hulkdemo17.17",
            "name": "HulkDemo17",
            "purgeTime": "",
            "qos": {
                "burstIOPS": 1500,
                "burstTime": 60,
                "curve": {
                    "4096": 100,
                    "8192": 160,
                    "16384": 270,
                    "32768": 500,
                    "65536": 1000,
                    "131072": 1950,
                    "262144": 3900,
                    "524288": 7600,
                    "1048576": 15000
                },
                "maxIOPS": 1000,
                "minIOPS": 100
            },
            "scsiEUIDeviceID": "306f746f00000011f47acc0100000000",
            "scsiNAADeviceID": "6f47acc100000000306f746f00000011",
            "sliceCount": 1,
            "status": "active",
            "totalSize": 53687091200,
            "virtualVolumeID": null,
            "volumeAccessGroups": [
                1
            ],
            "volumeID": 17,
            "volumePairs": []
        },
        {
            "access": "readWrite",
            "accountID": 1,
            "attributes": {},
            "blockSize": 4096,
            "createTime": "2016-06-23T14:19:18Z",
            "deleteTime": "",
            "enable512e": false,
```

```
                "iqn": "iqn.2010-01.com.solidfire:0oto.hulkdemo18.18",
                "name": "HulkDemo18",
                "purgeTime": "",
                "qos": {
                    "burstIOPS": 1500,
                    "burstTime": 60,
                    "curve": {
                        "4096": 100,
                        "8192": 160,
                        "16384": 270,
                        "32768": 500,
                        "65536": 1000,
                        "131072": 1950,
                        "262144": 3900,
                        "524288": 7600,
                        "1048576": 15000
                    },
                    "maxIOPS": 1000,
                    "minIOPS": 100
                },
                "scsiEUIDeviceID": "306f746f00000012f47acc0100000000",
                "scsiNAADeviceID": "6f47acc100000000306f746f00000012",
                "sliceCount": 1,
                "status": "active",
                "totalSize": 53687091200,
                "virtualVolumeID": null,
                "volumeAccessGroups": [
                    1
                ],
                "volumeID": 18,
                "volumePairs": []
            },
            {
                "access": "readWrite",
                "accountID": 1,
                "attributes": {},
                "blockSize": 4096,
                "createTime": "2016-06-24T15:21:59Z",
                "deleteTime": "",
                "enable512e": true,
                "iqn": "iqn.2010-01.com.solidfire:0oto.bk.24",
                "name": "BK",
                "purgeTime": "",
                "qos": {
                    "burstIOPS": 15000,
                    "burstTime": 60,
```

```
                    "curve": {
                        "4096": 100,
                        "8192": 160,
                        "16384": 270,
                        "32768": 500,
                        "65536": 1000,
                        "131072": 1950,
                        "262144": 3900,
                        "524288": 7600,
                        "1048576": 15000
                    },
                    "maxIOPS": 15000,
                    "minIOPS": 50
                },
                "scsiEUIDeviceID": "306f746f00000018f47acc0100000000",
                "scsiNAADeviceID": "6f47acc100000000306f746f00000018",
                "sliceCount": 1,
                "status": "active",
                "totalSize": 10737418240,
                "virtualVolumeID": null,
                "volumeAccessGroups": [],
                "volumeID": 24,
                "volumePairs": [
                    {
                        "clusterPairID": 2,
                        "remoteReplication": {
                            "mode": "Async",
                            "pauseLimit": 3145728000,
                            "remoteServiceID": 14,
                            "resumeDetails": "",
                            "snapshotReplication": {
                                "state": "Idle",
                                "stateDetails": ""
                            },
                            "state": "Active",
                            "stateDetails": ""
                        },
                        "remoteSliceID": 8,
                        "remoteVolumeID": 8,
                        "remoteVolumeName": "PairingDoc",
                        "volumePairUUID": "229fcbf3-2d35-4625-865a-
d04bb9455cef"
                    }
                ]
            }
        ]
```

```
    }
}
```

TestHardwareConfig

`TestHardwareConfig` 方法返回类似于以下示例的响应。

```
{
"id": 1,
"result": {
    "nodes": [
        {
            "nodeID": 1,
            "result": {
                "details": {
                    "BIOS_REVISION": {
                        "Passed": true,
                        "actual": "2.0",
                        "comparator": ">=",
                        "expected": "1.0.0.0"
                    },
                    "BIOS_VENDOR": {
                        "Passed": true,
                        "actual": "SolidFire",
                        "comparator": "==",
                        "expected": "SolidFire"
                    },
                    "BIOS_VERSION": {
                        "Passed": true,
                        "actual": "2.0.19",
                        "comparator": ">=",
                        "expected": "2.0.19"
                    },
                    "CPU_CORES_00": {
                        "Passed": true,
                        "actual": "6",
                        "comparator": "==",
                        "expected": "6"
                    },
                    "CPU_CORES_01": {
                        "Passed": true,
                        "actual": "6",
                        "comparator": "==",
                        "expected": "6"
                    },
```

```
                    "CPU_CORES_ENABLED_00": {
                        "Passed": true,
                        "actual": "6",
                        "comparator": "==",
                        "expected": "6"
                    },
                    "CPU_CORES_ENABLED_01": {
                        "Passed": true,
                        "actual": "6",
                        "comparator": "==",
                        "expected": "6"
                    },
                    "CPU_MODEL_00": {
                        "Passed": true,
                        "actual": "Intel(R) Xeon(R) CPU E5-2620 v2 @
2.10GHz",
                        "comparator": "==",
                        "expected": "Intel(R) Xeon(R) CPU E5-2620 v2 @
2.10GHz"
                    },
                    "CPU_MODEL_01": {
                        "Passed": true,
                        "actual": "Intel(R) Xeon(R) CPU E5-2620 v2 @
2.10GHz",
                        "comparator": "==",
                        "expected": "Intel(R) Xeon(R) CPU E5-2620 v2 @
2.10GHz"
                    },
                    "CPU_THREADS_00": {
                        "Passed": true,
                        "actual": "12",
                        "comparator": "==",
                        "expected": "12"
                    },
                    "CPU_THREADS_01": {
                        "Passed": true,
                        "actual": "12",
                        "comparator": "==",
                        "expected": "12"
                    },
                    "CPU_THREADS_ENABLED": {
                        "Passed": true,
                        "actual": "24",
                        "comparator": "==",
                        "expected": "24"
                    },
```

```
                    "IDRAC_VERSION": {
                        "Passed": true,
                        "actual": "2.41.40.40",
                        "comparator": ">=",
                        "expected": "1.06.06"
                    },
                    "MEMORY_GB": {
                        "Passed": true,
                        "actual": "64",
                        "comparator": ">=",
                        "expected": "64"
                    },
                    "MEMORY_MHZ_00": {
                        "Passed": true,
                        "actual": "1600",
                        "comparator": ">=",
                        "expected": "1333"
                    },
                    "MEMORY_MHZ_01": {
                        "Passed": true,
                        "actual": "1600",
                        "comparator": ">=",
                        "expected": "1333"
                    },
                    "MEMORY_MHZ_02": {
                        "Passed": true,
                        "actual": "1600",
                        "comparator": ">=",
                        "expected": "1333"
                    },
                    "MEMORY_MHZ_03": {
                        "Passed": true,
                        "actual": "1600",
                        "comparator": ">=",
                        "expected": "1333"
                    },
                    "MEMORY_MHZ_04": {
                        "Passed": true,
                        "actual": "1600",
                        "comparator": ">=",
                        "expected": "1333"
                    },
                    "MEMORY_MHZ_05": {
                        "Passed": true,
                        "actual": "1600",
                        "comparator": ">=",
```

```
                "expected": "1333"
            },
            "MEMORY_MHZ_06": {
                "Passed": true,
                "actual": "1600",
                "comparator": ">=",
                "expected": "1333"
            },
            "MEMORY_MHZ_07": {
                "Passed": true,
                "actual": "1600",
                "comparator": ">=",
                "expected": "1333"
            },
            "MPTSAS_BIOS_VERSION": {
                "Passed": true,
                "actual": "07.24.01.00",
                "comparator": "ANY",
                "expected": "7.25.0.0"
            },
            "MPTSAS_FIRMWARE_VERSION": {
                "Passed": true,
                "actual": "13.00.57.00",
                "comparator": "==",
                "expected": "13.0.57.0"
            },
            "NETWORK_DRIVER_ETH0": {
                "Passed": true,
                "actual": "bnx2x",
                "comparator": "==",
                "expected": "bnx2x"
            },
            "NETWORK_DRIVER_ETH1": {
                "Passed": true,
                "actual": "bnx2x",
                "comparator": "==",
                "expected": "bnx2x"
            },
            "NETWORK_DRIVER_ETH2": {
                "Passed": true,
                "actual": "bnx2x",
                "comparator": "==",
                "expected": "bnx2x"
            },
            "NETWORK_DRIVER_ETH3": {
                "Passed": true,
```

```
                        "actual": "bnx2x",
                        "comparator": "==",
                        "expected": "bnx2x"
                    },
                    "NETWORK_FIRMWARE_VERSION_ETH0": {
                        "Passed": true,
                        "actual": "7.10.18-solidfire-5f3ccbc781d53",
                        "comparator": "==",
                        "expected": "7.10.18-solidfire-5f3ccbc781d53"
                    },
                    "NETWORK_FIRMWARE_VERSION_ETH1": {
                        "Passed": true,
                        "actual": "7.10.18-solidfire-5f3ccbc781d53",
                        "comparator": "==",
                        "expected": "7.10.18-solidfire-5f3ccbc781d53"
                    },
                    "NETWORK_FIRMWARE_VERSION_ETH2": {
                        "Passed": true,
                        "actual": "7.10.18-solidfire-5f3ccbc781d53",
                        "comparator": "==",
                        "expected": "7.10.18-solidfire-5f3ccbc781d53"
                    },
                    "NETWORK_FIRMWARE_VERSION_ETH3": {
                        "Passed": true,
                        "actual": "7.10.18-solidfire-5f3ccbc781d53",
                        "comparator": "==",
                        "expected": "7.10.18-solidfire-5f3ccbc781d53"
                    },
                    "NUM_CPU": {
                        "Passed": true,
                        "actual": "2",
                        "comparator": "==",
                        "expected": "2"
                    },
                    "Parse failure in /var/log/sf-bios.info": {
                        "Passed": true,
                        "actual": "false",
                        "comparator": "==",
                        "expected": "false"
                    }
                },
                "duration": "00:00:00.195067",
                "result": "Passed"
            }
        }
    ]
```

```
  }
  }
```

Copyright Information

Copyright © 2022 NetApp, Inc. All rights reserved. Printed in the U.S. No part of this document covered by copyright may be reproduced in any form or by any means-graphic, electronic, or mechanical, including photocopying, recording, taping, or storage in an electronic retrieval system- without prior written permission of the copyright owner.

Software derived from copyrighted NetApp material is subject to the following license and disclaimer:

THIS SOFTWARE IS PROVIDED BY NETAPP “AS IS” AND WITHOUT ANY EXPRESS OR IMPLIED WARRANTIES, INCLUDING, BUT NOT LIMITED TO, THE IMPLIED WARRANTIES OF MERCHANTABILITY AND FITNESS FOR A PARTICULAR PURPOSE, WHICH ARE HEREBY DISCLAIMED. IN NO EVENT SHALL NETAPP BE LIABLE FOR ANY DIRECT, INDIRECT, INCIDENTAL, SPECIAL, EXEMPLARY, OR CONSEQUENTIAL DAMAGES (INCLUDING, BUT NOT LIMITED TO, PROCUREMENT OF SUBSTITUTE GOODS OR SERVICES; LOSS OF USE, DATA, OR PROFITS; OR BUSINESS INTERRUPTION) HOWEVER CAUSED AND ON ANY THEORY OF LIABILITY, WHETHER IN CONTRACT, STRICT LIABILITY, OR TORT (INCLUDING NEGLIGENCE OR OTHERWISE) ARISING IN ANY WAY OUT OF THE USE OF THIS SOFTWARE, EVEN IF ADVISED OF THE POSSIBILITY OF SUCH DAMAGE.

NetApp reserves the right to change any products described herein at any time, and without notice. NetApp assumes no responsibility or liability arising from the use of products described herein, except as expressly agreed to in writing by NetApp. The use or purchase of this product does not convey a license under any patent rights, trademark rights, or any other intellectual property rights of NetApp.

The product described in this manual may be protected by one or more U.S. patents, foreign patents, or pending applications.

RESTRICTED RIGHTS LEGEND: Use, duplication, or disclosure by the government is subject to restrictions as set forth in subparagraph (c)(1)(ii) of the Rights in Technical Data and Computer Software clause at DFARS 252.277-7103 (October 1988) and FAR 52-227-19 (June 1987).

Trademark Information

NETAPP, the NETAPP logo, and the marks listed at http://www.netapp.com/TM are trademarks of NetApp, Inc. Other company and product names may be trademarks of their respective owners.